Exercise

2026-10-05

Exercise 1. Let $R \subset T$ be domains with T finitely generated as a ring over R, and not algebraic over R. Prove that T is not a G-domain.

Exercise 2. Let R be a domain with quotient field K, and u a non-zero element of R. Let J be the kernel of the homomorphism μ from $R[x]$ into K given by $x \mapsto u^{-1}$. Prove: $J = (ux - 1)$.

Exercise 3. Let R be an integral domain having only a finite number of prime ideals. Prove that R is a G-domain.

Exercise 4. Let R be a Hilbert ring having only a finite number of maximal ideals. Prove that these are the only prime ideals in R.

Exercise 5. Let R be an integral domain with quotient field K. Assume that K is countably generated as a ring over R. Prove that R is a G-domain if and only if K cannot be expressed as the union of a properly ascending sequence of subrings containing R.

Exercise 6. Let K be any field (not necessarily algebraically closed). Prove that any maximal ideal in $K[x_1, \cdots, x_n]$ can be generated by n elements.

Exercise 7. Let R be a Hilbert ring such that every maximal ideal in R can be generated by k elements, k fixed. Prove that any maximal ideal in $R[x_1, \cdots, x_n]$ can be generated by $k + n$ elements.

Exercise 8. Let R be a Hilbert ring in which every maximal ideal is finitely generated. Prove that the same is true for $R[x]$.

Exercise 9. Prove that the following statements are equivalent:

(a) R is a Hilbert ring;

(b) Every radical ideal in R is an intersection of maximal ideals;

(c) Every prime ideal in R is an intersection of maximal ideals;

(d) Every G-ideal in R is an intersection of maximal ideals.

Exercise 10. Let R be any commutative ring. Let $\mathfrak{m}$ be a maximal ideal in $T = R[x_1, \cdots, x_n]$ satisfying $\mathfrak{m} \cap R = 0$. Prove that $\mathfrak{m}$ can be generated by $n + 1$ elements.

Exercise 11. Let K be algebraically closed and $R = K[x_1, \cdots, x_n]$. Let $f_1, \ldots, f_m \in R$. Then the equations

$$f_1 = 0, \quad f_2 = 0, \quad \ldots, \quad f_m = 0$$

have a simultaneous solution if and only if there do not exist $g_1, \ldots, g_m \in R$ with

$$\sum g_i f_i = 1.$$

Exercise 12. Let K be a field and write A for the vector space of n-tuples over K. Given an ideal $I \subset K[x_1, \cdots, x_n]$, we write $V(I)$ for the subset of A consisting of all points in A where all the polynomials in I vanish, and we call such a set a *variety*. Given a subset S of A, we write $J(S)$ for the ideal of polynomials vanishing on S. Assume that K is algebraically closed.

(1) Prove that $J(V(I))$ is the radical of I.

(2) Prove that the map $S \to J(S)$ sets up a one-to-one correspondence between all varieties and all radical ideals.

(3) Prove that under set-theoretic inclusion the set of radical ideals is a complete lattice and that the set of varieties likewise is a complete lattice. Prove further that the correspondence of part (2) is order-inverting.

(4) In the lattice of varieties of part (3), prove that the union of a finite number of varieties is just the set-theoretic union.

(5) Prove that in the correspondence of part (2) an ideal is prime if and only if the corresponding variety is *irreducible*, in the sense that it cannot be exhibited as the union of two properly smaller varieties.

Exercise 13. The homogeneous Nullstellensatz. K is a field, $R = K[x_1, \cdots, x_n]$. Call an ideal I in R *homogeneous* if along with a polynomial f, I contains the homogeneous constituents of f. Call a variety a *cone* if $(a_1, \cdots, a_n) \in V$ implies that $(ta_1, \cdots, ta_n) \in V$ for all $t \in K$.

(1) If I is homogeneous, prove that $V(I)$ is a cone.

(2) If V is a cone, and K is infinite, prove that $J(V)$ is homogeneous. Give an example to show that we need to assume that K is infinite.

(3) Thus or otherwise, prove that when K is algebraically closed, the radical of a homogeneous ideal is homogeneous. (This is actually true for any field.)

Exercise 14. Prove: an element x in a ring R is nilpotent if and only if it is a zero-divisor on every R-module.

Exercise 15. Let K be a field, L a field containing K, which is finitely generated as a ring over K. Prove that L is finite-dimensional over K.

Exercise 16. Let $R \subset T$ be rings with R a Hilbert ring and T finitely generated as a ring over R. Prove that any maximal ideal in T contracts to a maximal ideal in R.

Exercise 17. Let $R_1 \subset R_2 \subset R_3$ be rings with R_3 finitely generated as a ring over R_1. Let I_3 be an ideal in R_3 and I_1, I_2 its contractions to R_1, R_2. Prove: if I_1, I_3 are maximal, so is I_2.